**SE311 SPRING 2025-2026 / 04-03-2026 & 05-03-2026**
**Lab Work 02 – Iterator Pattern**

In this lab work, your task will require you to implement the Iterator pattern from scratch (in-class version). You are not allowed to use built-in Iterators in Java.

Your task is to write a program that traverses a collection of **Cars** using various filtering criteria.

1. **Create a `Car` class.** This class has three attributes: `model`, `engineType`, and `price`.
   - `model`: The name of the car (e.g., "Tesla Model 3").
   - `engineType`: The category of the engine (Electric, Gasoline, Hybrid, etc.).
   - `price`: Represents the cost of the car in USD.
   - Initialize the values through the Constructor.
   - Implement getter/setter only for `price` attribute.
   - override toString() method using the **format:** model \t engineType \t price.
2. **Make the necessary changes** to integrate the `Car` class into the code. When your newly modified code runs, it should print all the cars you add in the main. Use the following data in your main:
   - "Tesla Model 3", "Electric", 42000
   - "Fiat 500", "Gasoline", 18000
   - "Toyota Corolla", "Hybrid", 28000
   - "Porsche Taycan", "Electric", 90000
   - "Hyundai i20", "Gasoline", 15000
   - "Honda Civic", "Hybrid", 30000
3. **Add a new Iterator: `ElectricCarIterator`.** As the name suggests, it will traverse only the "Electric" type cars. Therefore, if you use the example data above, you should see 2 cars on the screen.
4. **Add another Iterator: `AffordableCarIterator`.** This iterator will traverse only the cars in which the price is equal to or under $30,000. Using the example data, you should see 4 cars on the screen.
5. **Implement a way to create these iterators** through a `createIterator()` method within your collection class.
6. **Run your completed code** and show the usage of 3 different iterators in main.